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$T(k)_s$

$T(k)_w$

$$= 3 \text{ Vol } \frac{\partial T}{\partial t}$$

COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

Department of Chemical Engineering

University of Illinois Chicago

Midsemester Exam #2

Only one sheet of paper 8 1/2 x 11 is allowed

No phones or i-pads or computers are allowed

Duration: 70 minutes --- October 21, 2025

All Problems must be done

(40)

$n r^2 \cdot h$

$\text{Vol} = A \cdot h$
 $A = \frac{\text{Vol}}{h}$

Problem 1: (50 points)

Let's design a shell and tube heat exchanger to heat a stream with $m_c C_{p,c} = 5,000 \text{ BTU} / \text{h}^\circ\text{F}$ from 90°F to 210°F flowing in the tubes using as the heat source a stream with $m_h C_{p,h} = 10,000 \text{ BTU} / \text{h}^\circ\text{F}$ entering the shell at 260°F . The heat exchanger will contain tubes with a 0.5 in. outside diameter. In order to maintain a reasonable fluid velocity inside the tubes, $N_t = 30$ tubes per pass will be used. The overall heat transfer coefficient is $U_o = 150 \text{ BTU} / \text{h ft}^2 \text{ }^\circ\text{F}$. For a heat exchanger with one-tube pass and one-shell pass, determine the required tube length. **(40 points)**

Suppose that we need to have shorter tubes, what can we do? **(10 points)**

$$D = 0.5 \text{ in} \quad | \quad r = 0.25 \text{ in}$$

$$N_t = 30 \text{ tubes/pass}$$

$$U = 150$$

$$A = \pi r^2 =$$

$$SA = 4\pi r^2 =$$

$$n \cdot q_1 = m_c c_p \Delta T = U A \Delta T_{lm} \cdot h$$

$$q_2 = m_c c_p \Delta T$$

10

Problem 2: (60 points)

$$\frac{10 \text{ mm}}{1000} = 0.01 \text{ m} = 1 \text{E-2}$$

$$T(t) = 40^\circ\text{C}$$

$$T_w = 20^\circ\text{C}$$

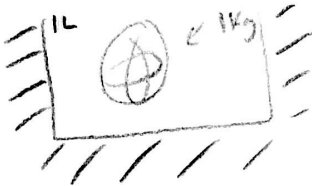
$$T_s = 90^\circ\text{C}$$

Let's consider a case where 1 kg of aluminum spheres with radius $R = 10 \text{ mm}$ initially at 90°C are immersed in 1 L of water initially at 20°C in a tank. The tank is insulated and heat losses are negligible. The water is assumed to be perfectly mixed. Determine the time needed to cool the aluminum spheres to 40°C assuming that the external heat transfer resistance at the sphere-fluid interface is dominant and that $h = 500 \text{ W / m}^2 \text{ K}$.

a) The density of aluminum is $\rho = 2,700 \text{ kg / m}^3$ and its heat capacity is $C_p = 900 \text{ J / kg K}$. Assume that the water temperature remains constant. (30 points)

b) If the changing temperature of the water is considered, determine the time needed to cool the aluminum spheres to 40°C . (30 points)

a) assume all spheres are one big sphere 20+5



$$\text{Vol} = \frac{m}{\rho} \quad \text{Vol} = 3.703 \text{E-4 m}^3$$

$$\text{Vol} = \frac{4}{3} \pi r^3$$

$$r = 0.044 \text{ m}$$

Want $T(t)$ so use lumped method

$$-h A_s (T - T_w) = \rho \text{Vol} C_p \frac{\partial T}{\partial t}$$

$$\rho = \frac{m}{\text{Vol}}$$

$$\text{Vol} = \frac{m}{\rho} = \frac{1 \text{ kg}}{2,700 \frac{\text{kg}}{\text{m}^3}} = 3.703 \text{E-4 m}^3$$

$$= m \cdot C_p \frac{\partial T}{\partial t}$$

$$= [1 \cdot C_p]$$

$$-h(4\pi r^2)(T - T_w) = C_p \frac{\partial T}{\partial t} \rightarrow \int_0^{t_F} \frac{-(h4\pi r^2)}{C_p} dt = \int_{T_1}^{T_2} \frac{1}{T - T_w} dT$$

$$\rightarrow \frac{-(h4\pi r^2)}{C_p} t_F = \ln(T - T_w) + A$$

$$t_F = \left(\frac{\ln(T_2 - T_w) C_p}{-(h4\pi r^2)} - \frac{\ln(T_1 - T_w) C_p}{-(h4\pi r^2)} \right) = \frac{-C_p}{h4\pi r^2} \ln \left(\frac{T_2 - T_w}{T_1 - T_w} \right)$$

$$t_F = 1312.48 \text{ s}$$

$$\sim 21.8 \text{ mins}$$

b) need $T_w(t)$ $\frac{dE}{dt} = E_{in} - E_{out}$

$C \rightarrow 9$

$$\rho_w \text{Vol}_w C_{pw} \frac{dT_w}{dt} = q_{\text{from sphere}} = m_s C_{ps} \Delta T_s = +45000 \text{ J}$$

assume q_r is

the same
from last problem

then

$$\int_{T_w}^{T_w} dT_w = \int \frac{m_s C_{ps} \Delta T_s}{\rho_w \text{Vol}_w C_{pw}} dt$$

$$T_w(t) - T_{w_0} = \frac{+t \cdot 45000 \text{ J}}{\rho_w \text{Vol}_w C_{pw}}$$

$$T_w(t) = \frac{+t(45000 \text{ J})}{\rho_w \text{Vol}_w C_{pw}} + T_{w_0} = \frac{+t(\text{J})}{\frac{1 \text{ kg}}{\text{L}} \cdot 1 \text{ L} \cdot 4180 \frac{\text{J}}{\text{kg} \cdot \text{K}}}$$

$\downarrow \quad \downarrow \quad \downarrow$
 $? \quad 1 \text{ L} \quad ? \quad 293.15$

$$\frac{1000 \frac{\text{kg}}{\text{m}^3} \cdot 1 \text{ m}^3}{1000 \text{ L}} \cdot 4180 \frac{\text{J}}{\text{kg} \cdot \text{K}} \cdot \frac{1000 \text{ J}}{1 \text{ kg}}$$

$$\frac{1 \text{ kg}}{\text{L}} \quad 4180 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

$$T_w(t) = +t \cdot 10.76 \frac{\text{K}}{\text{s}} \quad \text{back to a)} \quad -h(4\pi r^2)(T - t \cdot 10.76) = C_p \frac{dT}{dt}$$

$$\int_0^t h \cdot 10.76 \cdot 4\pi r^2 \cdot dt = \int_{T_1}^{T_2} \frac{C_p}{-h \cdot 4\pi r^2 T} dT$$

$$\frac{t^2}{2} h \cdot 10.76 \cdot 4\pi \cdot r^2 = \frac{C_p}{-h \cdot 4\pi r^2} (\ln(T_2) - \ln(T_1))$$

$\ln(T_2/T_1)$

$$t = \sqrt{\frac{2 C_p \ln(T_2/T_1)}{-h^2 4^2 \pi^2 r^4 \cdot 10.76}} = 0.046 \text{ s}$$